Problem Statement and Goals
Software Engineering

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Table 1 records revisions to this document.

Table 1: Revision History

Date	Developer(s)	Change
Sep 28th 2026	All	Revision 0

1 Problem Statement

This project addresses the need for accessible, individualized skill practice that provides autistic learners with meaningful activities and gives caregivers and educators interpretable information about their performance.

1.1 Problem

Autistic individuals differ in their strengths, communication styles, sensory preferences, and support needs. These needs can also change over time. Consequently, learning activities must accommodate individual differences rather than assume that the same challenge, presentation, or assistance will suit every learner. The World Health Organization identifies this diversity and the importance of accessible support [4].

For autistic learners seeking support with everyday routines, functional communication, or remembering task information, useful practice must balance a clear learning purpose with an understandable and engaging experience. Caregivers and educators need to select relevant activities, adjust the difficulty and

presentation, and account for the learner's preferences. An activity's educational purpose alone does not establish whether an individual can comfortably participate in it or wishes to repeat it.

Caregivers and educators also need to understand what changes in activity performance mean. A higher score may reflect greater familiarity with a task or reduced difficulty. Without information about these conditions, comparisons across sessions can be misleading. Research on the planning-assessment game ShopAutiPlan illustrates the value of examining how a task is completed alongside whether it is completed. However, interpreting game measures as evidence of a broader ability requires validation specific to the activity and intended population [3].

The problem addressed by this project is therefore the need to bring together individually appropriate practice, voluntary engagement, and contextualized performance information within a manageable caregiver-supported experience. The intended deliverable is a unified suite of learning games that allows caregivers and educators to configure practice and review a learner's activity history.

The initial scope comprises three activities addressing routine sequencing, functional communication, and short-term memory. These activities will provide opportunities to practise selected skills and observe performance under documented conditions. The project will not diagnose autism, assign an overall developmental level, or treat improved game performance as proof of improvement outside the activities.

1.2 Inputs and Outputs

The system will use information about the learner's selected objectives, preferences, and activity responses to provide individualized practice and contextualized results. Table 2 summarizes these inputs and outputs.

The outputs serve two purposes: supporting the learner during practice and helping the caregiver or educator make informed decisions about subsequent activities.

Table 2: High-level inputs and outputs

Inputs	Outputs
Learning objectives selected by the caregiver or educator with consideration of learner preferences	Practice activities aligned with the selected objectives
Learner accessibility preferences and selected difficulty levels	An activity experience reflecting the selected configuration
Learner responses and attempts	Immediate feedback and activity-specific performance results
Current and previous session outcomes, including the conditions under which they occurred	Individual performance histories and comparisons across comparable sessions
Caregiver or educator observations	Context accompanying recorded outcomes and supporting their interpretation

1.3 Stakeholders

Autistic learners are the primary users and beneficiaries. Their priorities include relevant activities, understandable instructions, comfortable interaction, and the ability to communicate preferences and stop participating.

Caregivers and educators, including parents, guardians, support workers, and teachers, administer activities and interpret results. They need to configure practice without extensive technical knowledge and understand performance in the context of the activity and difficulty selected.

Schools, learning centres, and support organizations are potential adopting organizations. Their interests include compatibility with available equipment, manageable staff effort, accessibility, and appropriate handling of learner information.

Domain advisers, including special-education professionals, communication specialists, and autism researchers, may inform activity selection and review the interpretation of results. Their role is advisory; no existing partnership is assumed.

Learner perspectives will be considered alongside caregiver and professional input. This approach reflects the importance of supported choice and communication emphasized by the Autistic Self Advocacy Network [1].

1.4 Environment

The initial target population is autistic children aged six to 12 participating with a caregiver or educator. This boundary limits the content and evaluation scope; age alone will not determine whether an activity is suitable for a particular learner.

The activities will support learners who can make intentional selections among visual choices through a supported input method, independently or with assistance accessing the controls. Independent reading and spoken responses will not be prerequisites. Caregivers may explain or demonstrate instructions.

Participation needs will be described explicitly rather than through a single "high-functioning" or "low-functioning" category. This recognizes that an individual's abilities and support needs can differ across tasks and settings [2].

The intended settings are homes, schools, and learning centres. The initial release will operate through a web browser on supported desktop, laptop, and tablet configurations, with internet access available during sessions. Interaction will accommodate keyboard, pointer, and touch input within the declared supported configurations. Sessions will allow caregiver assistance, breaks, and early stopping.

The initial scope excludes specialized input hardware, unsupervised multiplayer interaction, clinical assessment, and replacement of an established communication aid.

2 Goals

The following five goals define the project's core benefits and the evidence needed to demonstrate their achievement. Success will be evaluated through software verification, content review, and formative evaluation with intended users. These evaluations will establish whether the prototype meets its project goals; they will not establish clinical effectiveness.

2.1 Personalize practice to individual learning needs

Enable caregivers and educators to select an appropriate difficulty level for each learner and activity. Independent configuration across activities will accommodate differences in a learner's strengths and support needs.

Success criterion: Each of the three core activities provides at least three meaningful challenge levels. At least 80% of participating caregivers or educators can configure an activity for a specified learner scenario without developer assistance following a brief introduction.

2.2 Provide purposeful, repeatable skill practice

Provide three complete activities addressing routine sequencing, functional communication, and short-term memory. Each activity will offer repeated practice toward a clearly defined objective, with feedback that helps the learner understand the task and their response.

Success criterion: All three activities support a complete practice session and have a documented connection between their learning objective, task, feedback, and performance measures. Each activity's content and interpretation limits will be reviewed by a relevant domain adviser, with identified concerns addressed before final evaluation.

2.3 Offer an engaging experience that respects learner choice

Provide activities that intended learners find enjoyable and are willing to participate in, while preserving their ability to pause, decline, or stop. Engagement will be evaluated for each activity rather than inferred from completion scores or time spent playing.

Success criterion: For each core activity, at least 80% of participating learners who try it provide a positive enjoyment response through an accessible feedback method. Evaluation will also report voluntary continuation or replay choices, early stopping, and unavailable feedback so that the result does not conceal unsuccessful experiences.

Learner feedback and caregiver observations will be reported separately. Findings will be limited to the participants and conditions evaluated.

2.4 Make changes in performance understandable

Provide caregivers and educators with accurate activity histories that support decisions about whether to repeat, simplify, or advance practice. Results will identify the activity and difficulty associated with each session so that changes can be interpreted in context.

Success criterion: Reported results match expected outcomes across all defined verification cases. At least 80% of participating caregivers or educators correctly interpret at least 80% of representative progress examples, including examples involving changes in difficulty.

The product will report performance within defined activities rather than combine unrelated results into an overall developmental score.

2.5 Support accessible and comfortable participation

Allow learners to use the core activities through supported interaction methods and sensory settings. Accessibility will address how learners access an activity,

complementing the focus on learning challenge in Section 2.1.

Success criterion: Every core activity can be completed through each declared supported input method, with audio and nonessential motion disabled. Learners can pause or end activities, and no essential task depends exclusively on precise dragging, sound, or colour.

The interface will undergo review against applicable Web Content Accessibility Guidelines criteria and cognitive-accessibility guidance, with the evaluated scope and unresolved limitations documented. Familiar presentation and configurable interfaces are supported by cognitive-accessibility guidance from the World Wide Web Consortium [5].

Evaluation Basis

The 80% thresholds are proposed acceptance targets intended to require success for a clear majority while exposing unsuccessful experiences. They are not clinical benchmarks. Formative evaluation will aim to include at least five caregivers or educators and at least five intended learners per activity; participant counts and individual outcomes will be reported alongside percentages.

Detailed procedures will be specified in the verification and validation plan. Participation will depend on recruitment and applicable university requirements. If intended learners cannot participate, expert review may support design improvements, but the learner-engagement goal will remain unverified.

3 Stretch Goals

The following extensions will be considered after the core goals are met. They expand presentation options and opportunities for shared play.

3.1 Expand caregiver-controlled customization

Allow caregivers and educators to further customize the visual presentation of games to match individual learners' needs and preferences. Options will include in-game brightness, colour palettes, and the amount of decorative detail displayed. This extension will provide more control than the basic accessibility settings included in the core scope.

Success criterion: A caregiver or educator can adjust and save these presentation preferences for an individual learner, and the preferences are applied consistently across the core games. Supported combinations preserve readable instructions and distinguishable interactive elements.

3.2 Enable multiplayer interaction

Allow learners to participate in games with other players, creating opportunities for shared play and interaction. Caregivers or educators will be able to select

multiplayer activities appropriate to the learners' needs and preferences.

4 Custom Documents (CDs)

Two custom documents will record user needs and usability findings to support project decisions and evaluation.

4.1 Interview Report — Fall Semester

This report will summarize interviews with caregivers, educators, and, where feasible, autistic individuals to identify learning needs, accessibility preferences, and expectations for progress tracking. The interview report will explain how the findings inform the project scope and requirements, while acknowledging limitations in the perspectives represented.

4.2 Usability Evaluation Report — Winter Semester

This report will evaluate how intended users interact with the prototype, including caregiver configuration, interpretation of performance information, and learner participation in games. The usability evaluation report will document usability findings, resulting improvements, and evaluation limitations, with results linked to the project goals.

References

[1] Autistic Self Advocacy Network. Self-determination. https://autisticadvocacy.org/self-determination/. Accessed September 27, 2026.

[2] National Autistic Society. Varying support needs. https://www.autism.org.uk/advice-and-guidance/what-is-autism/varying-support-needs. Accessed September 27, 2026.

[3] Athmar N. M. Shamhan, Marrianna Constantinou, Howraa Hajaig, Marwa Qaraqe, and Dena Al-Thani. ShopAutiPlan: Validating a serious game for assessing executive planning in autism. *Autism Research*, 19(7):e70281, 2026. https://doi.org/10.1002/aur.70281.

[4] World Health Organization. Autism. https://www.who.int/news-room/fact-sheets/autism-spectrum-disorders, 2025. Accessed September 27, 2026.

[5] World Wide Web Consortium. Support a personalized and familiar interface. https://www.w3.org/WAI/WCAG2/supplemental/patterns/o8p04-interface/. Accessed September 27, 2026.

Appendix — Reflection

The following responses describe what the team learned while defining the problem, selecting goals, and setting the project scope.

1. What went well while writing this deliverable?

   Writing the problem statement helped our team clarify the need behind our proposed suite of games. We examined why existing learning and recreational experiences may not meet every autistic learner's needs, what role our application could serve, and what we could reasonably claim about its impact.

   Discussing learner diversity helped us question the assumption that one difficulty level, interaction style, or sensory environment would suit everyone. We defined success around meaningful choices and adaptation to individual needs. Research and discussion also helped us distinguish game performance from broader developmental outcomes.

   Establishing measurable success criteria turned broad ideas such as "accessible," "personalized," and "engaging" into goals we can evaluate. This clarified what we need to demonstrate by the end of the project.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   Our largest challenge was balancing ambition with a scope we could implement and evaluate. Early discussions covered many developmental areas, game types, sensory settings, input methods, and analytics. We narrowed these possibilities to a core set of activities and moved additional features into stretch goals.

   We also had to decide what the application could responsibly claim. Terms such as "improve," "develop," and "assess" can suggest outcomes our software alone cannot establish. We therefore framed the project as providing practice opportunities and reporting performance within the activities.

   Defining engagement required more than completion scores or time spent playing. We chose to consider accessible learner feedback, voluntary continuation or replay, early stopping, and unavailable feedback. Caregiver observations will be reported separately so they do not substitute for learners' own perspectives.

   Finally, we needed measurable goals before evaluation procedures were fully defined. We set preliminary acceptance targets and documented their limitations, recognizing that the targets may need refinement during verification and validation planning.

3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

We narrowed the core scope to three activities addressing routine sequencing, functional communication, and short-term memory. These activities provide enough variation to investigate a shared platform while leaving time for accessibility, learner choice, configuration, and evaluation.

We also defined personalization as caregiver and educator configuration rather than automatic decisions about what is best for a learner. This limits technical scope and avoids unsupported assumptions about individual needs.

Our understanding of complexity shifted from the number of features to the care needed to support different users, maintain accessibility, record performance context, and make results understandable. We chose to prioritize depth and validation over a larger collection of games.

Feedback from our supervisor, intended users, and Teaching Assistants (TAs) may lead to further scope changes. Our current goals leave room for that feedback while defining a substantial project we can evaluate.